COMPUTER NETWORKING

Complete Bible — 2026

Beginner · Intermediate · Advanced · Pro/Expert

OSI & TCP/IP · Routing · Switching · DNS · HTTP · TLS

BGP · SDN · Cloud Networking · Wi-Fi 6/7 · 5G · Zero Trust

WHAT'S INSIDE:

40 fully-detailed chapters across 4 mastery levels

Complete protocol walkthroughs with packet diagrams

Real-world troubleshooting, Wireshark, and lab exercises

Generated: 2026

CONTENTS AT A GLANCE

QUICK REFERENCE

PART I

BEGINNER

For those starting from zero — no prior networking knowledge assumed

Chapter 1

What Is a Computer Network?

The Big Picture

A **computer network** is a group of devices connected together so they can share data and resources. Every time you send a text message, stream a video, or browse the web, your device is communicating over a computer network. The internet itself is simply the largest computer network on Earth — a network of networks connecting billions of devices.

Why Networks Exist

Purpose	Without Network	With Network
File sharing	Copy to USB, physically deliver	Instantly send over network
Communication	Phone calls only, no video	Email, chat, video calls, VoIP
Resource sharing	One printer per computer	100 computers share one printer
Centralized data	Data on each machine separately	Database accessible from anywhere
Internet access	Not possible	Every device can reach the web

Key Terminology

- **Host / End device**: Any device that sends or receives data — your laptop, phone, server, IoT sensor.
- **Client**: A device that requests services (e.g., your browser). **Server**: A device that provides services (e.g., web server).
- **Protocol**: A set of rules for communication. Like languages — both sides must speak the same one. Example: HTTP for web pages, SMTP for email.
- **Bandwidth**: Maximum data rate (e.g., 100 Mbps). **Latency**: Delay for data to travel from source to destination (e.g., 20 ms). **Throughput**: Actual data rate achieved.
- **Packet**: Data is broken into small chunks called packets. Each packet has a header (addressing info) and payload (actual data). Packets may take different paths and are reassembled at the destination.

Client-Server vs Peer-to-Peer

Aspect	Client-Server	Peer-to-Peer (P2P)
Architecture	Dedicated servers, many clients	All nodes are equal — both client and server
Examples	Web, email, databases	BitTorrent, Bitcoin, LAN file sharing
Scalability	Scale by adding servers	Scale by adding peers
Management	Centralized, easier to secure	Decentralized, harder to manage
Cost	Requires server hardware	Uses existing devices

Chapter 2

Network Types: LAN, WAN, MAN, PAN

Classification by Geographic Scope

Type	Full Name	Range	Speed	Example
PAN	Personal Area Network	~10 m	1-3 Mbps	Bluetooth headphones
LAN	Local Area Network	Building/campus	1-100 Gbps	Office network, home Wi-Fi
MAN	Metropolitan Area Network	City	10-100 Gbps	Cable TV network, city fiber
WAN	Wide Area Network	Country/global	Varies	The Internet, corporate MPLS
SAN	Storage Area Network	Data center	16-128 Gbps	Fibre Channel to storage arrays

The Internet

The **Internet** is a global WAN — a network of networks. It connects millions of LANs, MANs, and other networks using standardized protocols (TCP/IP). No single organization owns the Internet. Internet Service Providers (ISPs) connect end users to the network. ISPs are organized in tiers: **Tier 1** (backbone providers like AT&T;, NTT — they peer freely with each other), **Tier 2** (regional ISPs that pay Tier 1 for transit and peer with each other), **Tier 3** (local ISPs that buy transit from Tier 2).

Intranet and Extranet

An **intranet** is a private network within an organization (uses internet protocols but not accessible from outside). An **extranet** extends the intranet to trusted external partners (e.g., suppliers accessing your inventory system via VPN).

Chapter 3

Network Topologies and Hardware

Physical Topologies

Topology	Description	Pros	Cons
Bus	All devices on a single cable	Simple, cheap	Single point of failure, collisions
Star	All devices connect to central switch	Easy to manage, isolate faults	Switch failure takes down network
Ring	Each device connects to two neighbors	Equal access, deterministic	One break disrupts ring
Mesh	Every device connects to every other	Highly redundant, fault-tolerant	Expensive, complex cabling
Hybrid	Combination (e.g., star-bus)	Flexible, scalable	Complex design

Network Hardware

Device	OSI Layer	Function	Intelligence
Hub	Layer 1 (Physical)	Broadcasts frames to all ports	None (dumb repeater)
Switch	Layer 2 (Data Link)	Forwards frames by MAC address	Learns MAC-port mappings
Router	Layer 3 (Network)	Forwards packets by IP address between networks	Routing tables, protocols
Access Point	Layer 1-2	Connects wireless clients to wired network	SSID, encryption
Firewall	Layer 3-7	Filters traffic by rules	Stateful inspection, DPI
Load Balancer	Layer 4-7	Distributes traffic across servers	Health checks, algorithms

Cables and Media

Media	Max Speed	Max Distance	Use Case
Cat5e (UTP)	1 Gbps	100 m	Office LANs
Cat6a (UTP)	10 Gbps	100 m	Data centers, high-speed LANs
Single-mode fiber	100+ Gbps	80+ km	WAN links, data center interconnects
Multi-mode fiber	100 Gbps	~550 m	Short data center runs
Coaxial	1 Gbps	500 m	Cable internet (DOCSIS)

Chapter 4

The OSI Model — All 7 Layers

Why Layering?

Networking is complex. Layering breaks the problem into manageable pieces. Each layer has a specific job and communicates with the layers directly above and below it. This lets us change one layer (e.g., switch from Ethernet to Wi-Fi) without affecting the others.

The 7 Layers

#	Layer	Function	PDU	Protocols/Examples
7	Application	User interface, app services	Data	HTTP, FTP, SMTP, DNS, SSH
6	Presentation	Data formatting, encryption	Data	SSL/TLS, JPEG, ASCII, MPEG
5	Session	Session management, dialogs	Data	NetBIOS, RPC, PPTP
4	Transport	End-to-end delivery, reliability	Segment	TCP, UDP, QUIC
3	Network	Logical addressing, routing	Packet	IP, ICMP, OSPF, BGP
2	Data Link	Physical addressing, framing	Frame	Ethernet, Wi-Fi (802.11), PPP
1	Physical	Bits on wire/air	Bits	Cables, connectors, voltages

Memory aid: 'Please Do Not Throw Sausage Pizza Away' (Physical → Application)

Encapsulation

As data flows down the layers, each layer adds its own header (and sometimes trailer). This is **encapsulation**. At the sender: Application data → Transport adds TCP header (segment) → Network adds IP header (packet) → Data Link adds Ethernet header + trailer (frame) → Physical converts to bits. At the receiver, each layer strips its header — this is **de-encapsulation**.

```
// Data encapsulation:
[Application Data]
[TCP Header | Application Data]          <- Segment
[IP Header | TCP Header | App Data]      <- Packet
[Eth Header | IP | TCP | App Data | FCS]  <- Frame
[10110011011010...]                       <- Bits on wire
```

Chapter 5

The TCP/IP Model

TCP/IP vs OSI

The **TCP/IP model** (also called the Internet model) is the practical model used on the Internet. The OSI model is a theoretical reference. TCP/IP has 4 layers:

TCP/IP Layer	OSI Equivalent	Key Protocols
Application	Layers 5-7 (Session, Presentation, Application)	HTTP, DNS, SMTP, FTP, SSH, TLS
Transport	Layer 4 (Transport)	TCP, UDP, QUIC
Internet	Layer 3 (Network)	IPv4, IPv6, ICMP, ARP, IGMP
Network Access	Layers 1-2 (Physical, Data Link)	Ethernet, Wi-Fi, PPP

How Data Flows

When you type a URL in your browser: (1) **Application**: Browser creates HTTP request. (2) **Transport**: TCP breaks it into segments, adds port numbers (source: random, dest: 80/443). (3) **Internet**: IP adds source and destination IP addresses, determines next hop. (4) **Network Access**: Ethernet adds MAC addresses, converts to electrical signals. At each router: strip data link header → examine IP header → make routing decision → add new data link header → send on next link. At destination: reverse the process up the layers.

The IP Protocol

IP (Internet Protocol) is the heart of the Internet. It provides: **addressing** (every device gets a unique IP address), **routing** (packets find their way across networks), and **fragmentation** (break large packets into smaller ones if needed). IP is **connectionless** (no setup, each packet independent) and **best-effort** (no guarantee of delivery — TCP handles reliability).

Chapter 6

IP Addressing and Subnetting

IPv4 Address Structure

An IPv4 address is 32 bits, written as 4 decimal octets: e.g., **192.168.1.100**. Total addresses: 2^32 = ~4.3 billion. An IP address has two parts: **network portion** (identifies the network) and **host portion** (identifies the device on that network). The **subnet mask** defines where the split is.

```
// IP address: 192.168.1.100 with mask 255.255.255.0 (/24)
// Binary:
// IP:   11000000.10101000.00000001.01100100
// Mask: 11111111.11111111.11111111.00000000
//       |--- Network (24 bits) ---| Host(8)|
// Network: 192.168.1.0    Host part: .100
// Usable hosts: 2^8 - 2 = 254 (minus network + broadcast)
```

Private vs Public Addresses

Range	CIDR	Class	# Addresses	Use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	A	16.7 million	Large enterprises
172.16.0.0 – 172.31.255.255	172.16.0.0/12	B	1 million	Medium networks
192.168.0.0 – 192.168.255.255	192.168.0.0/16	C	65,536	Home/small office

Subnetting

Subnetting divides a network into smaller sub-networks. This improves security (isolate departments), reduces broadcast domains, and uses addresses efficiently. To subnet: borrow bits from the host portion. Each borrowed bit doubles the number of subnets but halves the hosts.

```
// Subnet 192.168.1.0/24 into 4 subnets:
// Borrow 2 bits: /24 -> /26 (255.255.255.192)
// Subnet 1: 192.168.1.0/26   (hosts .1-.62,   bcast .63)
// Subnet 2: 192.168.1.64/26  (hosts .65-.126,  bcast .127)
// Subnet 3: 192.168.1.128/26 (hosts .129-.190, bcast .191)
// Subnet 4: 192.168.1.192/26 (hosts .193-.254, bcast .255)
// Each subnet: 2^6 - 2 = 62 usable hosts
```

Chapter 7

Ethernet and Switching Basics

Ethernet (IEEE 802.3)

Ethernet is the dominant LAN technology. It operates at **Layer 2** using MAC addresses (48-bit, e.g., AA:BB:CC:DD:EE:FF). Ethernet defines how data is framed, addressed, and transmitted on a shared medium.

Ethernet Frame Format

```
// Ethernet II frame:
[Preamble 7B][SFD 1B][Dest MAC 6B][Src MAC 6B][Type 2B]
[Payload 46-1500B][FCS 4B]
// Type: 0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6
// FCS: Frame Check Sequence (CRC-32 for error detection)
// MTU: Maximum Transmission Unit = 1500 bytes (payload)
```

How Switches Work

A switch builds a **MAC address table** (CAM table) that maps MAC addresses to ports. When a frame arrives: (1) Learn: record source MAC → ingress port. (2) Lookup: find destination MAC in table. (3) If found: forward to that port only (**unicast**). If not found: **flood** to all ports except ingress. This is far more efficient than a hub (which always floods).

MAC Address	Port	Age
AA:BB:CC:11:22:33	Fa0/1	10s
AA:BB:CC:44:55:66	Fa0/3	5s
AA:BB:CC:77:88:99	Fa0/7	2s

Full Duplex and Auto-Negotiation

Half duplex: Can send OR receive, not both (old hubs). **Full duplex**: Can send AND receive simultaneously (modern switches). **Auto-negotiation**: Devices automatically agree on speed (10/100/1000 Mbps) and duplex. Mismatched settings cause performance issues — always verify.

Chapter 8

Routing Fundamentals

What Is Routing?

Routing is the process of finding the best path for a packet from source to destination across multiple networks. Routers operate at **Layer 3** and use IP addresses to make forwarding decisions. Each router has a **routing table** that lists known networks and next hops.

Static vs Dynamic Routing

Aspect	Static Routing	Dynamic Routing
Configuration	Manual (admin enters routes)	Automatic (protocol discovers routes)
Scalability	Poor (manual updates)	Good (adapts automatically)
CPU/Memory	Low (no protocol overhead)	Higher (runs routing protocol)
Use case	Small networks, default routes	Medium to large networks
Examples	ip route 10.0.0.0/8 192.168.1.1	OSPF, EIGRP, BGP, RIP

The Routing Table

```
// Example routing table output:
Destination      Mask             Gateway          Interface  Metric
10.0.0.0         255.0.0.0        192.168.1.1      eth0       10
192.168.2.0      255.255.255.0    192.168.1.2      eth1       1
0.0.0.0          0.0.0.0          192.168.1.254    eth0       100
// 0.0.0.0/0 = default route (gateway of last resort)
```

How a Router Forwards a Packet

(1) Receive frame on interface. (2) Strip Layer 2 header. (3) Examine destination IP. (4) Consult routing table — find longest prefix match. (5) Determine next-hop IP and exit interface. (6) Build new Layer 2 header (ARP for next-hop MAC). (7) Send frame out exit interface. Note: the IP header stays the same (mostly), but the Ethernet header changes at every hop.

Chapter 9

TCP and UDP — Transport Layer

TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless (fire and forget)
Reliability	Guaranteed delivery, ordering, retransmission	No guarantees
Flow control	Yes (sliding window)	No
Congestion control	Yes (slow start, AIMD)	No
Header size	20-60 bytes	8 bytes
Speed	Slower (overhead)	Faster (minimal overhead)
Use cases	Web (HTTP), email, file transfer	DNS, VoIP, video streaming, gaming

TCP Three-Way Handshake

```
// TCP connection setup:
Client -> Server: SYN (seq=100)
Server -> Client: SYN-ACK (seq=300, ack=101)
Client -> Server: ACK (seq=101, ack=301)
// Connection established! Data can flow.
// TCP connection teardown (4-way):
Client -> Server: FIN
Server -> Client: ACK
Server -> Client: FIN
Client -> Server: ACK
```

Port Numbers

Port	Protocol	Service
20-21	TCP	FTP (data/control)
22	TCP	SSH
25	TCP	SMTP (email sending)
53	TCP/UDP	DNS
80	TCP	HTTP
443	TCP	HTTPS (TLS)
3389	TCP	RDP (Remote Desktop)

Chapter 10

DNS, DHCP, and NAT

DNS (Domain Name System)

DNS translates human-readable domain names (www.example.com) into IP addresses (93.184.216.34). Without DNS, you would have to memorize IP addresses for every website.

```
// DNS resolution for www.example.com:
1. Browser checks local cache
2. OS checks /etc/hosts file
3. Query recursive resolver (usually ISP's DNS)
4. Resolver asks Root server -> 'Go ask .com TLD server'
5. Resolver asks .com TLD -> 'Go ask example.com NS'
6. Resolver asks example.com authoritative NS
7. Gets answer: www.example.com = 93.184.216.34
8. Resolver caches answer (TTL) and returns to browser
```

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses to devices. Without DHCP, you would manually configure every device. The DHCP process is called **DORA**: **D**iscover (client broadcasts 'any DHCP servers?'), **O**ffer (server offers an IP), **R**equest (client requests that IP), **A**cknowledge (server confirms). DHCP also provides subnet mask, default gateway, and DNS server addresses.

NAT (Network Address Translation)

NAT allows multiple devices on a private network to share a single public IP address. Your home router does this: your devices have private IPs (192.168.x.x), but all outgoing traffic uses one public IP. The router maintains a **NAT table** mapping (private IP:port ↔ public IP:port). This was invented because IPv4 addresses ran out — NAT lets millions of devices hide behind one public IP.

PART II

INTERMEDIATE

For practitioners deepening their skills

Chapter 11

HTTP, HTTPS, and the World Wide Web

HTTP (HyperText Transfer Protocol)

HTTP is the protocol of the web. It is a **request-response** protocol: the client sends a request, the server sends a response. HTTP is **stateless** — each request is independent (cookies and sessions add state). HTTP/1.1 uses persistent connections (keep-alive). HTTP/2 adds multiplexing (multiple requests on one connection). HTTP/3 uses QUIC (UDP-based) for faster connections.

Method	Purpose	Idempotent?
GET	Retrieve resource	Yes
POST	Create resource / submit data	No
PUT	Replace resource entirely	Yes
PATCH	Partially update resource	No
DELETE	Remove resource	Yes
HEAD	Like GET but response has no body	Yes

HTTPS and TLS

HTTPS = HTTP over TLS (Transport Layer Security). TLS provides: **confidentiality** (encryption — no one can read the data), **integrity** (no one can modify data in transit), and **authentication** (server proves its identity via certificate). The TLS handshake: client sends supported ciphers → server picks a cipher and sends certificate → client verifies certificate, generates session key → encrypted communication begins.

HTTP Status Codes

Code	Meaning	Example
200	OK — request succeeded	Page loaded successfully
301	Moved Permanently — use new URL	Site migration redirect
404	Not Found — resource doesn't exist	Typo in URL
500	Internal Server Error — server bug	Application crash
503	Service Unavailable — overloaded	Server under maintenance

Chapter 12

VLANs and Inter-VLAN Routing

What Is a VLAN?

A **VLAN** (Virtual LAN) logically segments a physical switch into multiple broadcast domains. Devices in different VLANs cannot communicate without a router, even if they are on the same switch. This improves security (isolate departments), performance (smaller broadcast domains), and flexibility (move users by config, not cable).

802.1Q Trunking

A **trunk** link carries traffic for multiple VLANs between switches. IEEE **802.1Q** adds a 4-byte tag to the Ethernet frame containing the VLAN ID (12 bits = 4094 VLANs max). The **native VLAN** is untagged (default: VLAN 1). Access ports belong to one VLAN; trunk ports carry multiple VLANs.

Inter-VLAN Routing

Three methods: (1) **Router-on-a-stick**: Single router interface with subinterfaces, one per VLAN. Simple but bandwidth-limited. (2) **Layer 3 switch**: Switch with routing capability — creates SVI (Switch Virtual Interface) per VLAN. Fast, uses hardware routing. (3) **External router** with separate interfaces per VLAN (impractical for many VLANs).

Chapter 13

Spanning Tree Protocol (STP)

The Loop Problem

Redundant links in a switched network create **loops**. Broadcast frames circulate endlessly, consuming all bandwidth (**broadcast storm**). STP prevents this by logically blocking redundant paths, creating a loop-free tree topology while keeping backup paths available for failover.

STP Operation

(1) Elect a **Root Bridge** (lowest Bridge ID = priority + MAC). (2) Each non-root switch finds its **Root Port** (lowest cost path to root). (3) Each network segment gets a **Designated Port** (lowest cost to root on that segment). (4) All other ports are **Blocked**. Port states: Blocking → Listening → Learning → Forwarding. Convergence time: ~30-50 seconds (classic STP).

RSTP (Rapid STP, 802.1w)

RSTP converges in ~1-3 seconds (vs 30-50s for STP). It defines new port roles: **Alternate** (backup root port) and **Backup** (backup designated port). RSTP is backward-compatible with STP. Modern networks should always use RSTP or MSTP (Multiple Spanning Tree).

Chapter 14

Dynamic Routing: OSPF and EIGRP

Routing Protocol Types

Type	Algorithm	Examples	Scope
Distance Vector	Bellman-Ford	RIP, EIGRP	Share routing tables with neighbors
Link State	Dijkstra (SPF)	OSPF, IS-IS	Flood link-state info to all routers
Path Vector	Best path by attributes	BGP	Inter-AS routing (Internet)

OSPF (Open Shortest Path First)

OSPF is the most widely used interior routing protocol. It uses Dijkstra's algorithm to find shortest paths. Key concepts: **Areas** (divide network into manageable regions — Area 0 is the backbone), **LSA (Link-State Advertisement)** (each router floods info about its links), **Cost metric** (based on bandwidth: cost = reference BW / link BW). OSPF converges fast and scales well with areas.

EIGRP

Cisco's proprietary (now open) protocol. Uses DUAL algorithm. Metrics: bandwidth, delay, reliability, load. Very fast convergence (**feasible successors** provide instant backup routes). Sends updates only when changes occur (not periodic). Good for Cisco shops but less universal than OSPF.

Chapter 15

IPv6 Addressing and Transition

Why IPv6?

IPv4 has ~4.3 billion addresses — exhausted in 2011. IPv6 uses 128-bit addresses: $2^{128} = 3.4 \times 10^{38}$ addresses (enough for every grain of sand on Earth). Format: eight groups of 4 hex digits separated by colons, e.g., **2001:0db8:85a3:0000:0000:8a2e:0370:7334**. Leading zeros can be omitted; one group of consecutive zeros replaced by ::.

IPv6 Features

- **No NAT needed**: Every device gets a globally unique address.
- **No broadcast**: Uses multicast and anycast instead.
- **Simplified header**: Fixed 40 bytes (IPv4: 20-60 bytes variable). Faster router processing.
- **Auto-configuration (SLAAC)**: Devices generate their own addresses using router advertisements.
- **IPsec built-in**: Encryption support is mandatory (optional in IPv4).

Transition Mechanisms

Mechanism	Description
Dual Stack	Run IPv4 and IPv6 simultaneously on all devices
Tunneling (6to4, Teredo)	Encapsulate IPv6 inside IPv4 packets
NAT64/DNS64	Translate between IPv4 and IPv6 at gateway

Chapter 16

Wireless Networking: Wi-Fi Standards

Wi-Fi Standards (IEEE 802.11)

Standard	Name	Frequency	Max Speed	Year
802.11a	Wi-Fi 2	5 GHz	54 Mbps	1999
802.11n	Wi-Fi 4	2.4/5 GHz	600 Mbps	2009
802.11ac	Wi-Fi 5	5 GHz	6.9 Gbps	2014
802.11ax	Wi-Fi 6/6E	2.4/5/6 GHz	9.6 Gbps	2020
802.11be	Wi-Fi 7	2.4/5/6 GHz	46 Gbps	2024

Key Wireless Concepts

- **SSID**: Network name broadcast by the access point.
- **Authentication**: WPA3 (current best), WPA2 (still common), WEP (broken, never use).
- **Channels**: 2.4 GHz has 3 non-overlapping channels (1, 6, 11). 5 GHz has 24+ non-overlapping channels.
- **MIMO**: Multiple-Input Multiple-Output — use multiple antennas for higher throughput.
- **OFDMA (Wi-Fi 6+)**: Divide channel into sub-channels to serve multiple clients simultaneously.

Chapter 17

Network Security Fundamentals

CIA Triad

The foundation of information security: **Confidentiality** (only authorized users can read data — encryption), **Integrity** (data is not tampered with — hashing, checksums), **Availability** (systems are accessible when needed — redundancy, DDoS protection).

Common Attacks

Attack	Description	Defense
Man-in-the-Middle	Intercept communication between two parties	TLS, certificate pinning
ARP Spoofing	Fake ARP replies to redirect traffic	Dynamic ARP Inspection (DAI)
DNS Spoofing	Return fake DNS responses	DNSSEC
DDoS	Overwhelm target with traffic	Rate limiting, CDN, scrubbing
Phishing	Trick users into revealing credentials	Training, MFA, email filtering
SQL Injection	Inject malicious SQL via input fields	Parameterized queries, WAF

Chapter 18

Firewalls, ACLs, and VPNs

Firewall Types

Type	How It Works	Layer
Packet Filtering	Match rules on IP/port headers	Layer 3-4
Stateful Inspection	Track connection state + packet filtering	Layer 3-4
Application Gateway (Proxy)	Inspect application-layer content	Layer 7
Next-Gen Firewall (NGFW)	DPI + IDS/IPS + application awareness	Layer 3-7

Access Control Lists (ACLs)

ACLs are ordered lists of permit/deny rules applied to router interfaces. **Standard ACLs**: filter by source IP only. **Extended ACLs**: filter by source/destination IP, protocol, port. Rules are processed top-down; first match wins. Always end with an implicit deny all.

VPN (Virtual Private Network)

A VPN creates an encrypted tunnel over a public network (the Internet) to connect remote sites or users securely. Types: **Site-to-site** (IPsec — connect two offices), **Remote access** (SSL/TLS VPN — employee connects from home), **WireGuard** (modern, fast, simple).

Chapter 19

Network Address Translation In Depth

NAT Types

Type	Description	Use Case
Static NAT	1:1 mapping (private IP ↔ public IP)	Servers needing fixed public IP
Dynamic NAT	Pool of public IPs, assigned on demand	Multiple servers, limited public IPs
PAT (Port Address Translation)	Many:1 (all share one public IP, differentiated by port)	Home/office internet (most common)
NAT64	Translates IPv6 to IPv4	IPv6-only networks reaching IPv4 services

NAT Traversal Challenges

NAT breaks end-to-end connectivity: external hosts cannot initiate connections to internal hosts. Solutions: **Port forwarding** (manually map external port to internal IP:port), **UPnP/NAT-PMP** (automatic port mapping), **STUN/TURN/ICE** (used by WebRTC, VoIP to discover external address and relay traffic). NAT also complicates protocols that embed IP addresses in payloads (FTP, SIP).

Chapter 20

Network Troubleshooting and Tools

Troubleshooting Methodology

Follow a systematic approach: (1) Identify the problem (symptoms, scope). (2) Establish a theory (layer-by-layer — start at Physical or Application). (3) Test the theory. (4) Establish a plan. (5) Implement the fix. (6) Verify and document.

Essential Tools

Tool	Purpose	Example
ping	Test reachability (ICMP echo)	ping 8.8.8.8
traceroute/tracert	Show path packets take (each hop)	traceroute google.com
nslookup/dig	Query DNS records	dig A example.com
netstat/ss	Show active connections and ports	ss -tulnp
tcpdump	Capture packets on CLI	tcpdump -i eth0 port 80
Wireshark	GUI packet capture and analysis	Filter: http.request
nmap	Network scanning and port discovery	nmap -sV 192.168.1.0/24
iperf3	Test bandwidth between two hosts	iperf3 -s / iperf3 -c server

PART III

ADVANCED

For those seeking mastery of network internals

Chapter 21

TCP Internals: Congestion Control and Flow

TCP Flow Control

The receiver advertises a **receive window** (rwnd) — how much data it can buffer. The sender never sends more than rwnd bytes without acknowledgment. This prevents overwhelming a slow receiver.

TCP Congestion Control

The network can also be a bottleneck. TCP maintains a **congestion window** (cwnd) that limits how much unacknowledged data is in flight. Effective window = min(cwnd, rwnd). Algorithms:

Phase/Algorithm	Behavior
Slow Start	cwnd doubles every RTT (exponential growth) until ssthresh
Congestion Avoidance	cwnd increases by 1 MSS per RTT (linear growth)
Fast Retransmit	3 duplicate ACKs → retransmit lost segment immediately
Fast Recovery	After fast retransmit, set ssthresh=cwnd/2, cwnd=ssthresh
CUBIC (Linux default)	Cubic function for cwnd growth — faster recovery after loss
BBR (Google)	Model bandwidth and RTT to find optimal sending rate

TCP Optimizations

- **Selective ACK (SACK)**: Receiver reports which segments received — sender retransmits only missing ones.
- **Window Scaling**: Extend receive window beyond 64 KB (up to 1 GB) for high-bandwidth links.
- **TCP Fast Open (TFO)**: Send data in the SYN packet to eliminate one RTT on reconnection.
- **Nagle's Algorithm**: Buffer small writes and send as one segment. Disable (TCP_NODELAY) for low-latency apps.

Chapter 22

BGP — The Internet's Routing Protocol

What Is BGP?

BGP (Border Gateway Protocol) is the routing protocol that holds the Internet together. It enables ~75,000+ autonomous systems (ASes) to exchange routing information. BGP is a **path-vector** protocol: each route includes the full AS path it traversed. BGP runs over TCP port 179.

eBGP vs iBGP

Aspect	eBGP (External)	iBGP (Internal)
Between	Different ASes	Routers within same AS
TTL	1 (directly connected)	255 (multi-hop)
Next-hop	Changes at each eBGP hop	Preserved (must resolve via IGP)
Full mesh required?	No	Yes (or use route reflectors)

BGP Path Selection

When BGP has multiple paths to a destination, it selects the best using these criteria (in order): (1) Highest **Weight** (Cisco-specific, local), (2) Highest **Local Preference**, (3) Locally originated route, (4) Shortest **AS Path**, (5) Lowest **Origin** type (IGP < EGP < incomplete), (6) Lowest **MED**, (7) eBGP over iBGP, (8) Lowest IGP cost to next hop, (9) Oldest route, (10) Lowest Router ID.

Route Reflectors and Confederations

iBGP requires a full mesh (every router peers with every other). For N routers, that is N(N-1)/2 sessions — impractical at scale. **Route reflectors** solve this: one router reflects routes to all iBGP peers, eliminating the full mesh. **Confederations** split a large AS into sub-ASes that use eBGP rules internally.

Chapter 23

MPLS and Traffic Engineering

MPLS (Multi-Protocol Label Switching)

MPLS inserts a **label** (32-bit) between Layer 2 and Layer 3 headers. Routers (called **Label Switch Routers**) forward packets based on the label — much faster than IP lookup. At the network edge, an **ingress LER** pushes a label; at the core, **LSRs** swap labels; at the egress, the **egress LER** pops the label. MPLS creates **Label Switched Paths (LSPs)** — pre-determined paths through the network.

MPLS Applications

- **L3 VPN (VRF)**: Service providers create isolated virtual routing tables per customer using MPLS.
- **L2 VPN (VPLS, EVPN)**: Extend Layer 2 domains across the WAN — sites appear to be on the same LAN.
- **Traffic Engineering**: Steer traffic along specific paths to avoid congestion (RSVP-TE, SR-MPLS).

Segment Routing

Segment Routing (SR) is replacing traditional MPLS signaling (LDP, RSVP-TE). Instead of per-hop signaling, the ingress router encodes the entire path as a stack of segment IDs (labels). This eliminates complex signaling protocols and is well-suited for SDN controllers.

Chapter 24

TLS/SSL Cryptography and PKI

TLS 1.3 Handshake

TLS 1.3 (2018) is the current standard. It reduces the handshake from 2 round-trips (TLS 1.2) to **1 round-trip**: Client sends: supported ciphers + key share. Server responds: chosen cipher + key share + certificate + encrypted data. Both sides compute shared secret. Total: 1-RTT to first data (0-RTT for resumption).

Certificate Authority (CA) and PKI

A **PKI** (Public Key Infrastructure) enables trust. A **Certificate Authority** (e.g., Let's Encrypt, DigiCert) signs a server's public key, producing a **certificate**. Your browser trusts a set of root CAs. When connecting to a server, the browser verifies the certificate chain up to a trusted root. If valid, the server is authenticated.

Cipher Suites

A cipher suite specifies: key exchange (ECDHE), authentication (RSA/ECDSA), bulk encryption (AES-256-GCM, ChaCha20), and hash (SHA-384). TLS 1.3 removed insecure options: no RSA key exchange, no CBC mode, no SHA-1. Only 5 cipher suites remain.

Chapter 25

Software-Defined Networking (SDN)

SDN Architecture

Traditional networking: control plane (routing decisions) and data plane (forwarding) are tightly coupled on each device. **SDN** separates them: a centralized **SDN controller** makes all routing decisions and programs the forwarding tables on switches. Benefits: centralized management, programmability, vendor-agnostic, faster innovation.

SDN Layers

- **Application Layer**: Network apps (load balancers, firewalls, monitoring) via northbound API (REST).
- **Control Layer**: SDN controller (OpenDaylight, ONOS, Cisco ACI) — the 'brain' of the network.
- **Infrastructure Layer**: Physical/virtual switches programmed via southbound API (OpenFlow, P4, gRPC).

OpenFlow

OpenFlow is the original SDN protocol. The controller installs **flow rules** in switches: match fields (src/dst IP, port, VLAN) → actions (forward, drop, modify). Each packet is matched against the flow table; if no match, sent to controller for decision. While pioneering, OpenFlow is being supplemented by more modern approaches (P4, gRPC/gNMI).

Chapter 26

Network Function Virtualization (NFV)

What Is NFV?

NFV replaces dedicated hardware appliances (firewalls, load balancers, WAN optimizers) with **Virtual Network Functions (VNFs)** running on standard servers. Benefits: reduced hardware costs, faster deployment, elastic scaling. NFV is complementary to SDN — SDN handles forwarding decisions, NFV handles network services.

NFV Architecture

- **NFVI** (Infrastructure): Physical compute, storage, network + virtualization layer (hypervisor/containers).
- **VNFs**: Software implementations of network functions (virtual firewall, virtual router).
- **MANO** (Management & Orchestration): Lifecycle management of VNFs — deploy, scale, terminate.

Chapter 27

Load Balancing and High Availability

Load Balancing Methods

Method	How It Works	Best For
Round Robin	Distribute requests sequentially	Homogeneous servers
Least Connections	Send to server with fewest active connections	Long-lived connections
Weighted	Assign weight based on server capacity	Heterogeneous servers
IP Hash	Hash source IP to determine server	Session persistence
Least Response Time	Send to fastest-responding server	Latency-sensitive apps

L4 vs L7 Load Balancing

L4: Operates on IP/port — fast, simple, no content inspection (e.g., TCP connection routing). **L7**: Inspects HTTP headers, cookies, URL paths — can route /api to one pool, /images to another. L7 is more flexible but more resource-intensive. Modern LBs: HAProxy, NGINX, AWS ALB/NLB, Envoy.

High Availability

HA ensures services survive failures. Techniques: **Active-Passive** (standby takes over on failure — VRRP, HSRP), **Active-Active** (both nodes serve traffic — load balanced), **DNS failover** (change DNS record on failure), **Anycast** (multiple servers share same IP — routing selects nearest).

Chapter 28

QoS — Quality of Service

Why QoS?

Not all traffic is equal. Voice/video need low latency and jitter; file downloads can tolerate delay. QoS prioritizes critical traffic over best-effort traffic, especially when links are congested.

QoS Mechanisms

Mechanism	Description
Classification	Mark packets with priority (DSCP, CoS)
Policing	Drop/remark traffic exceeding a rate limit
Shaping	Buffer excess traffic and send at a controlled rate
Queuing	Weighted Fair Queuing (WFQ), Priority Queuing, CBWFQ
Congestion Avoidance	WRED — randomly drop packets before queue fills

DSCP (Differentiated Services Code Point)

DSCP uses 6 bits in the IP header (64 values). Common markings: **EF** (Expedited Forwarding, 46) for voice, **AF** classes (Assured Forwarding) for important data, **BE** (Best Effort, 0) for everything else. Per-Hop Behaviors (PHBs) define how routers treat each DSCP value.

Chapter 29

Multicast and IGMP

Unicast vs Broadcast vs Multicast

Type	Source	Destination	Use Case
Unicast	One	One	Web browsing, email
Broadcast	One	All on network	ARP, DHCP discover
Multicast	One	Group of interested receivers	IPTV, video conferencing, stock feeds

Multicast is efficient: one stream serves many receivers. The sender sends once; routers replicate only where receivers exist. Multicast uses IP range 224.0.0.0 - 239.255.255.255.

IGMP (Internet Group Management Protocol)

Hosts use IGMP to tell their local router which multicast groups they want to join. Routers use PIM (Protocol Independent Multicast) to build multicast distribution trees across the network.

Chapter 30

Network Monitoring: SNMP, NetFlow, sFlow

Monitoring Approaches

Tool	Type	What It Provides
SNMP	Pull-based polling	Device metrics (CPU, memory, interface counters)
NetFlow/IPFIX	Flow-based export	Who is talking to whom, how much data, which apps
sFlow	Sampled packets	Traffic visibility with minimal overhead
SNMP Traps	Push-based alerts	Device sends notification on events (link down, etc.)
Syslog	Log messages	Event logs from devices (errors, auth, config changes)

SNMP (Simple Network Management Protocol)

SNMP uses a **manager** (monitoring server) and **agents** (on each device). The manager polls agents for metrics stored in a **MIB** (Management Information Base) — a tree of OIDs (Object Identifiers). SNMP v3 adds authentication and encryption (v1/v2c use community strings — essentially plaintext passwords). Tools: Zabbix, PRTG, Nagios, LibreNMS.

PART IV

PRO / EXPERT

Staff-level judgment and craft

Chapter 31

Cloud Networking: AWS, Azure, GCP VPC

Virtual Private Cloud (VPC)

A **VPC** is an isolated virtual network in the cloud. You define IP ranges (CIDR), subnets (public/private), route tables, and internet gateways. VPCs are the networking foundation for all cloud services.

Key Cloud Networking Concepts

Concept	Description
Subnets	Divide VPC into public (internet-facing) and private (internal) subnets
Security Groups	Stateful L3-L4 firewall rules attached to instances (allow rules only)
NACLs	Stateless subnet-level firewall (allow + deny rules)
Internet Gateway	Connects VPC to the public internet
NAT Gateway	Lets private instances reach the internet (outbound only)
VPC Peering	Direct connection between two VPCs (no transitive routing)
Transit Gateway	Hub-and-spoke for connecting many VPCs and on-prem
PrivateLink	Access cloud services without traversing the internet

Hybrid Connectivity

VPN: Encrypted tunnel over internet (IPsec) — quick to set up, variable latency. **Direct Connect** (AWS) / **ExpressRoute** (Azure) / **Cloud Interconnect** (GCP): Dedicated physical link to cloud — predictable latency, higher bandwidth, more expensive.

Chapter 32

Container Networking: Docker, Kubernetes, CNI

Docker Networking

Docker creates a virtual bridge (**docker0**) and connects containers via **veth pairs**. Each container gets its own network namespace with a private IP. Traffic to the host or internet goes through NAT. Docker network modes: **bridge** (default, isolated), **host** (share host's network stack), **none** (no networking), **overlay** (multi-host — uses VXLAN).

Kubernetes Networking Model

Kubernetes requires: (1) Every pod gets a unique IP. (2) Pods can communicate without NAT. (3) Nodes can communicate with pods without NAT. This is implemented by **CNI plugins**: Calico (L3 routing, network policies), Cilium (eBPF-based, L7 policies), Flannel (simple overlay), AWS VPC CNI (pods get VPC IPs). **Services** provide stable virtual IPs with load balancing. **Ingress** routes external HTTP(S) traffic to services.

Service Mesh

A **service mesh** (Istio, Linkerd) adds a sidecar proxy to every pod. This proxy handles: mTLS (mutual authentication), load balancing, retries, circuit breaking, and observability — all without changing application code. The mesh provides L7 traffic management across microservices.

Chapter 33

Zero Trust Architecture

The Principle

Traditional security: trust everything inside the network perimeter ('castle and moat'). **Zero Trust**: never trust, always verify — regardless of location. Every request is authenticated, authorized, and encrypted. There is no implicit trust based on network position.

Zero Trust Pillars

- **Identity**: Strong authentication (MFA), identity provider (Okta, Azure AD), least-privilege access.
- **Device**: Verify device health, posture, and compliance before granting access.
- **Network**: Micro-segmentation, encrypted connections (mTLS), no flat networks.
- **Application**: Application-level access control (BeyondCorp, Zscaler), not network-level VPN.
- **Data**: Classify, encrypt, and monitor all sensitive data at rest and in transit.

Implementation

Google's **BeyondCorp** pioneered zero trust: employees access internal apps via identity-aware proxies, not VPNs. The network location is irrelevant — an employee on the corporate LAN gets the same scrutiny as one at a coffee shop. Technologies: ZTNA (Zero Trust Network Access), SDP (Software-Defined Perimeter), SASE (Secure Access Service Edge).

Chapter 34

5G and Mobile Core Networks

5G Architecture

5G is not just 'faster 4G'. It introduces a fundamentally new architecture: (1) **eMBB** (Enhanced Mobile Broadband): 20 Gbps peak, 100 Mbps average. (2) **URLLC** (Ultra-Reliable Low-Latency): 1 ms latency for autonomous vehicles, remote surgery. (3) **mMTC** (Massive Machine-Type Communication): 1 million devices per km^2 for IoT.

5G Network Slicing

Network slicing creates multiple virtual networks on shared physical infrastructure. Each slice is optimized for a use case: a gaming slice prioritizes low latency, an IoT slice prioritizes device density, a video slice prioritizes bandwidth. Slices are isolated and can have independent security policies. This is enabled by NFV and SDN.

5G Core (5GC)

The 5G core is a cloud-native, microservices-based architecture. Key functions: AMF (Access and Mobility Management), SMF (Session Management), UPF (User Plane Function), PCF (Policy Control), UDM (Unified Data Management). All communicate via HTTP/2 and REST APIs — a radical departure from the 4G EPC's diameter protocol.

Chapter 35

Internet Exchange Points and Peering

What Is an IXP?

An **Internet Exchange Point** is a physical location where ISPs, CDNs, and content providers connect to exchange traffic directly — rather than routing through transit providers. Benefits: lower latency (shorter path), lower cost (peering is usually free), higher bandwidth. Major IXPs: DE-CIX (Frankfurt, 14+ Tbps peak), AMS-IX (Amsterdam), LINX (London).

Peering vs Transit

Aspect	Peering	Transit
Cost	Usually free (settlement-free)	Paid (customer pays provider)
Relationship	Equal exchange between peers	Customer buys full internet access
Traffic	Only between the two peers' networks	Full internet routing table
Where	At IXP or private interconnect	Via provider's network

Chapter 36

DDoS Mitigation and Anycast

DDoS Attack Types

Type	Layer	Example	Mitigation
Volumetric	L3-4	UDP flood, DNS amplification	Scrubbing centers, rate limiting
Protocol	L3-4	SYN flood, Ping of Death	SYN cookies, stateful inspection
Application	L7	HTTP flood, Slowloris	WAF, rate limiting, CAPTCHA

Anycast

Anycast: Multiple servers in different locations share the same IP address. BGP routing directs each client to the nearest server. Benefits: automatic geographic load distribution, DDoS absorption (attack traffic is distributed across all locations). Used by DNS root servers, CDNs (Cloudflare, Akamai), and DDoS scrubbing services.

Chapter 37

Network Automation: Ansible, Terraform, NETCONF

Why Automate?

Manual CLI configuration does not scale. Errors multiply. Changes take hours. **Network automation** treats network configuration as code: version-controlled, tested, repeatable. Infrastructure as Code (IaC) for networking.

Tools

Tool	Type	Protocol	Use Case
Ansible	Config management	SSH/NETCONF	Push configs to devices, templating
Terraform	Infrastructure as Code	API	Provision cloud networking (VPC, LB)
NETCONF	Config protocol	SSH (XML/YANG)	Standards-based device configuration
RESTCONF	Config protocol	HTTPS (JSON/YANG)	REST API for network devices
gNMI	Telemetry + config	gRPC	Streaming telemetry, model-driven config
Nornir	Python automation	SSH/API	Pythonic network automation framework

YANG Data Models

YANG is a data modeling language for network configuration. It defines the structure of config data (like a schema). OpenConfig YANG models are vendor-neutral, enabling the same automation to work across Cisco, Juniper, Arista. YANG models are used with NETCONF, RESTCONF, and gNMI.

Chapter 38

Observability: eBPF, OpenTelemetry, Distributed Tracing

Network Observability

Modern networks need more than SNMP polling. **Observability** = metrics + logs + traces. **eBPF** (extended Berkeley Packet Filter) lets you run custom programs in the kernel to capture packet-level data without modifying applications. Cilium Hubble uses eBPF to provide network flow visibility in Kubernetes.

Distributed Tracing

In microservices, a single request may traverse 10+ services. **Distributed tracing** follows the request across all services, measuring latency at each hop. Standards: OpenTelemetry (OTEL), Jaeger, Zipkin. Each request gets a trace ID; spans record timing at each service.

Streaming Telemetry

Instead of polling (SNMP), devices push data in real-time via **gNMI** (gRPC Network Management Interface). Sub-second granularity. Subscribe to specific YANG paths (e.g., interface counters, BGP neighbor state). Tools: Telegraf + InfluxDB + Grafana, Prometheus + Grafana.

Chapter 39

QUIC, HTTP/3, and the Future of Transport

QUIC

QUIC is Google's transport protocol, now standardized as RFC 9000. It runs over **UDP** and provides: (1) **Built-in TLS 1.3** — encryption is not optional. (2) **0-RTT connection setup** — first packet can carry data. (3) **Stream multiplexing without head-of-line blocking** — one lost packet does not block other streams (unlike TCP). (4) **Connection migration** — connections survive IP changes (e.g., Wi-Fi to cellular).

HTTP/3

HTTP/3 = HTTP over QUIC. All major browsers and CDNs support it. HTTP/3 eliminates TCP's head-of-line blocking problem, reduces connection setup time, and provides better performance on lossy networks (mobile, satellite). Adoption is growing rapidly — Cloudflare reports 30%+ of their traffic uses HTTP/3.

The Future

- **Multipath QUIC**: Use multiple network paths simultaneously (Wi-Fi + cellular).
- **MASQUE**: Proxying over QUIC — next-generation VPN and proxy protocols.
- **WebTransport**: Low-latency bidirectional communication for web apps over QUIC.

Chapter 40

Network Design Patterns and Trade-offs

Fundamental Design Trade-offs

Trade-off	Option A	Option B	Guidance
Flat vs Hierarchical	Simple, fewer devices	Scalable, isolate failures	Hierarchical for > 50 devices
L2 vs L3 at access	Simple, VLAN spanning	Better fault isolation, routing	L3 to access emerging standard
Underlay vs Overlay	Direct routing, simple	Flexible, multi-tenant	Overlay for cloud/multi-tenant
Hardware vs Software LB	High throughput	Flexible, cheaper	Software for most, HW for extreme scale
Centralized vs Distributed FW	Single policy point	Less bottleneck	Distributed (micro-seg) for zero trust

Modern Data Center Design

The **Clos/Spine-Leaf** topology replaced the old 3-tier (core/aggregation/access) model. Every leaf switch connects to every spine switch. Benefits: equal-cost multipath (ECMP), predictable latency, easy horizontal scaling. Add capacity by adding more spines. This is the standard in hyperscale data centers (Google, Facebook, Microsoft).

Design Principles

- **Design for failure**: Every component will fail. Ensure redundancy at every layer (dual power, dual uplinks, multi-path routing).
- **Automate everything**: Manual config is the #1 cause of outages. Use IaC, CI/CD for network changes.
- **Encrypt by default**: TLS everywhere, mTLS between services, WPA3 for wireless.
- **Monitor before you need to**: Deploy observability from day one. You can't fix what you can't see.
- **Keep it simple**: Every feature you enable is a feature that can break. Complexity is the enemy of reliability.

QUICK REFERENCE

Decision guides, cheat sheets, and at-a-glance comparisons

Quick Reference

Networking Cheat Sheet

OSI Layers at a Glance

```
L7 Application   - HTTP, DNS, SMTP, SSH, FTP
L6 Presentation  - TLS/SSL, JPEG, MPEG, compression
L5 Session        - RPC, PPTP, session management
L4 Transport      - TCP (reliable), UDP (fast), QUIC
L3 Network        - IP, ICMP, OSPF, BGP, ARP
L2 Data Link      - Ethernet, Wi-Fi, MAC addresses
L1 Physical       - Cables, connectors, signals, bits
```

Common Port Numbers

Port	Protocol	Port	Protocol
20-21	FTP	143	IMAP
22	SSH	161-162	SNMP
23	Telnet	443	HTTPS
25	SMTP	514	Syslog
53	DNS	3306	MySQL
67-68	DHCP	3389	RDP
80	HTTP	5060	SIP
110	POP3	8080	HTTP alt

Subnetting Quick Reference

```
CIDR  Mask             Hosts    Networks(/24 from /16)
/8    255.0.0.0       16.7M     1
/16   255.255.0.0     65,534    256
/24   255.255.255.0   254       65,536
/25   255.255.255.128 126       2 subnets per /24
/26   255.255.255.192 62        4 subnets per /24
/27   255.255.255.224 30        8 subnets per /24
/28   255.255.255.240 14        16 subnets per /24
/30   255.255.255.252 2         point-to-point links
/32   255.255.255.255 1         host route
```

Troubleshooting Commands

```
ping 8.8.8.8                # Test connectivity
traceroute google.com       # Trace packet path
dig A example.com           # Query DNS A record
curl -I https://example.com # HTTP headers only
ss -tulnp                   # Show listening ports
ip addr show                # Show IP addresses
ip route show               # Show routing table
tcpdump -i eth0 port 443    # Capture TLS traffic
nmap -sV 192.168.1.0/24    # Scan network services
```

```
iperf3 -c server -t 10     # Bandwidth test 10s
```

Key Formulas

```
Throughput = Window Size / RTT
Usable hosts per subnet = 2^(32-prefix) - 2
Number of subnets = 2^(new_bits_borrowed)
Bandwidth-Delay Product = Bandwidth * RTT
  (optimal TCP window size for full link utilization)
```